

Mariam Helal

📍 Berkeley, CA ✉ mariam.helal@berkeley.edu 📞 (626) 620-1627 🌐 marihelal 📱 marihelal

Education

University of California, Berkeley

Berkeley, CA, USA

BA in Planetary Science, BA in Astrophysics, Minor in Chemistry

May 2028

- GPA: 4.0/4.0
- Relevant Coursework: Introductory Astrophysics, General Chemistry, Organic Chemistry, Physical Chemistry, Planetary Astrophysics, Calculus, Differential Equations, Linear Algebra, General Physics, Physics of Planetary Interiors, Electricity & Magnetism

Mission Early College High School

Santa Clara, CA, USA

High School Diploma, AS-T in Physics

May 2024

Research

Ramsden Research Fellow

Berkeley, CA, USA

Macdonald Lab

Oct 2024 – present

- Developing protocol for usage of hand-held XRF instrument to analyze and interpret major element compositions of carbonates
- Investigating effects of diagenesis and mineralogy on $\delta^{13}\text{C}$ values of carbonates recording the Trezona anomaly

Undergraduate Researcher

Berkeley, CA, USA

Bergner Lab

Feb 2025 – present

- Modeling edge-on protoplanetary disks using radmc3d to analyze relationship between inclination and column density calculations
- Modeling ice emission in protoplanetary disks
- Conducted molecule identification and spectral analysis of ALMA proto-stars to probe phosphorus chemistry in planet-forming regions

Teaching

ASTRON/EPS C12 - Planets Reader

Berkeley, CA, USA

UC Berkeley Department of Astronomy & Department of Earth & Planetary Sciences

Jan 2025 – present

ASTRON/EPS C162 - Planetary Astrophysics Reader

Berkeley, CA, USA

UC Berkeley Department of Astronomy & Department of Earth & Planetary Sciences

Aug 2025 – Dec 2025

Python DeCal Student Facilitator

Berkeley, CA, USA

UC Berkeley Department of Astronomy

Jan 2025 – present

- Taught lectures on Python programming, plotting, and data analysis techniques
- Hosted office hours to provide students with individual support and address coding questions

ASTRON/EPS W12 - Planets Tutor/Reader

Remote

UC Berkeley Department of Astronomy & Department of Earth & Planetary Sciences

June 2025 – Aug 2025

- Interacted with students via discussions and office hours
- Point of contact for sections of the class, assisting student with material-related and administrative questions

Professional Experience

Chief Scientist

Remote

NASA L'SPACE Mission Concept Academy

Jan 2025 – May 2025

- Participated in weekly seminars covering the process of NASA mission concept development and learning from industry professionals

- Led team of 5 through science-delegated tasks, developing a NASA mission concept for a Aerobot system to conduct research on the Venusian atmosphere and surface and navigating surprise roadblocks through reevaluating priorities and descopeing
- Presented final PDR document to board of mentors, providing a summary of our mission and answering clarifying questions

Extracurriculars

Mentor	<i>Berkeley, CA, USA</i>
<i>Undergraduate Laboratory at Berkeley (ULAB), Physics & Astronomy</i>	<i>Sept 2025 – present</i>
◦ Mentored new undergraduates through research project analyzing ALMA observations of protoplanetary disks	
Telescope Crew Officer	<i>Berkeley, CA, USA</i>
<i>Undergraduate Astronomy Society</i>	<i>May 2025 – present</i>
◦ Operated & trained telescope crew in usage of Newtonian & Schmidt-Cassegrain telescopes	
◦ Trained in and operated the Richard Treffers 17" PlaneWave telescope, of the UC Berkeley Department of Astronomy	
◦ Co-facilitated weekly Sidewalk Astronomy outreach events to engage the campus community in public observation sessions	
MPS Scholars Mentor	<i>Berkeley, CA, USA</i>
<i>Math & Physical Sciences Department, UC Berkeley</i>	<i>May 2025 – present</i>
◦ Advised freshman on course selection, research opportunities, and academic planning within the EPS and Planetary Science disciplines	
◦ Hosted regular mentoring meetings to address student questions and provide academic and personal support	
Extrusion Sub-team Member	<i>Berkeley, CA, USA</i>
<i>Student Environmental Resource Center</i>	<i>Sept 2024 – May 2025</i>
◦ Researched and tested methods for extrusion of recycled 3D printer filament to create a closed-loop campus filament recycling program	
Sensor Placement Co-Lead	<i>Berkeley, CA, USA</i>
<i>Strawberry Creek Monitoring Group</i>	<i>Jan 2025 – Aug 2025</i>
◦ Created social media posts and graphic design materials for outreach on SCMG activities and creek health	
◦ Co-led sensor placement efforts, finding optimal locations to place sensors monitoring creek activity	

Awards & Honors

Ramsden Research Fellowship	<i>2025</i>
◦ Summer 2025 - \$7500	
◦ Fall 2025 - \$2000	
Dean's List	<i>Fall 2024, Fall 2025</i>
Presidential Volunteer Service Certificate	<i>Fall 2023</i>

Memberships

Geological Society of America	<i>2025</i>
<i>Student Member</i>	
American Physical Society	<i>2025</i>
<i>Student Member</i>	

Presentations

Evaluating the Handheld XRF as a Means for the Major Element Analysis of Carbonates	<i>Apr 2026</i>
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Helal, Mariam, Renger, Ellery, Tasistro-Hart, Adrian, Macdonald, Francis
2026 GSA Cordilleran Section Meeting

Modeling Water Ice in Protoplanetary Disks

Apr 2026

Helal, Mariam

UC Berkeley Physics & Astrophysics Undergraduate Research Colloquium

Skills

Lab: Microscopy, XRF, Mineral Separation

Coding: Python, Matplotlib, Pandas, NumPy, SciPy, Seaborn, Cartopy, LaTeX, HTML, CSS, HPC (Savio)

Programs: CASSIS, CARTA, radmc3d

Languages: English, Arabic